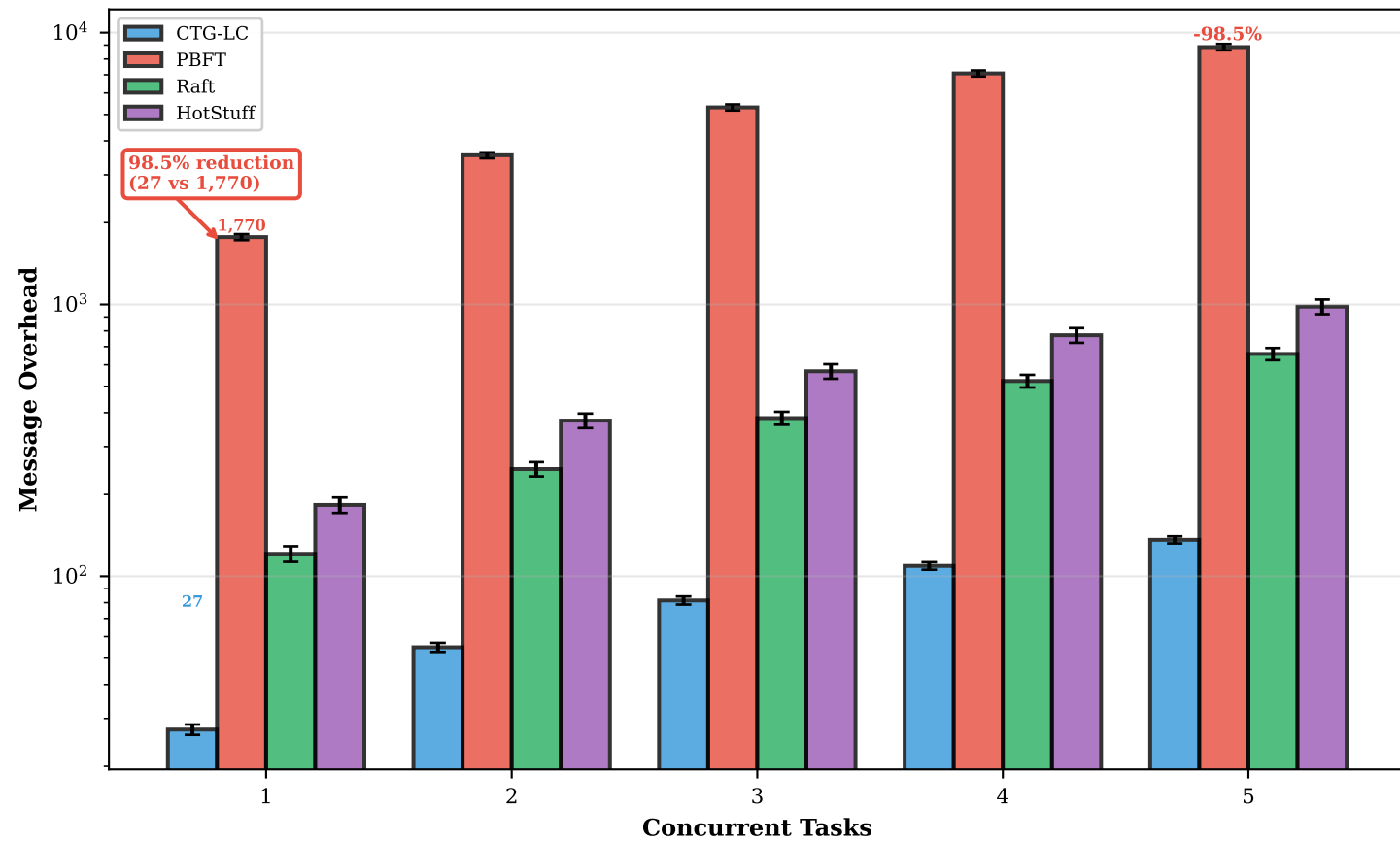
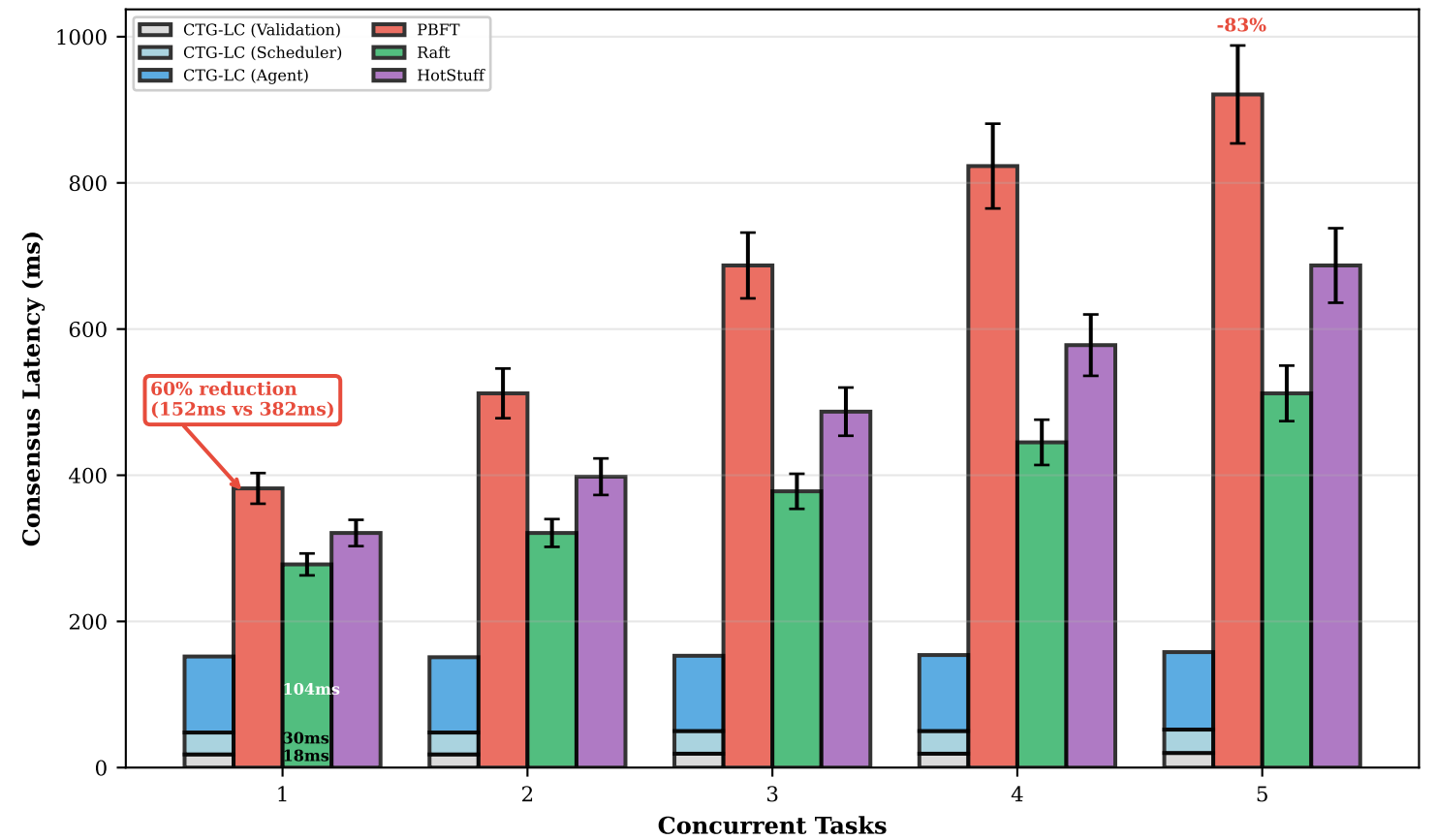


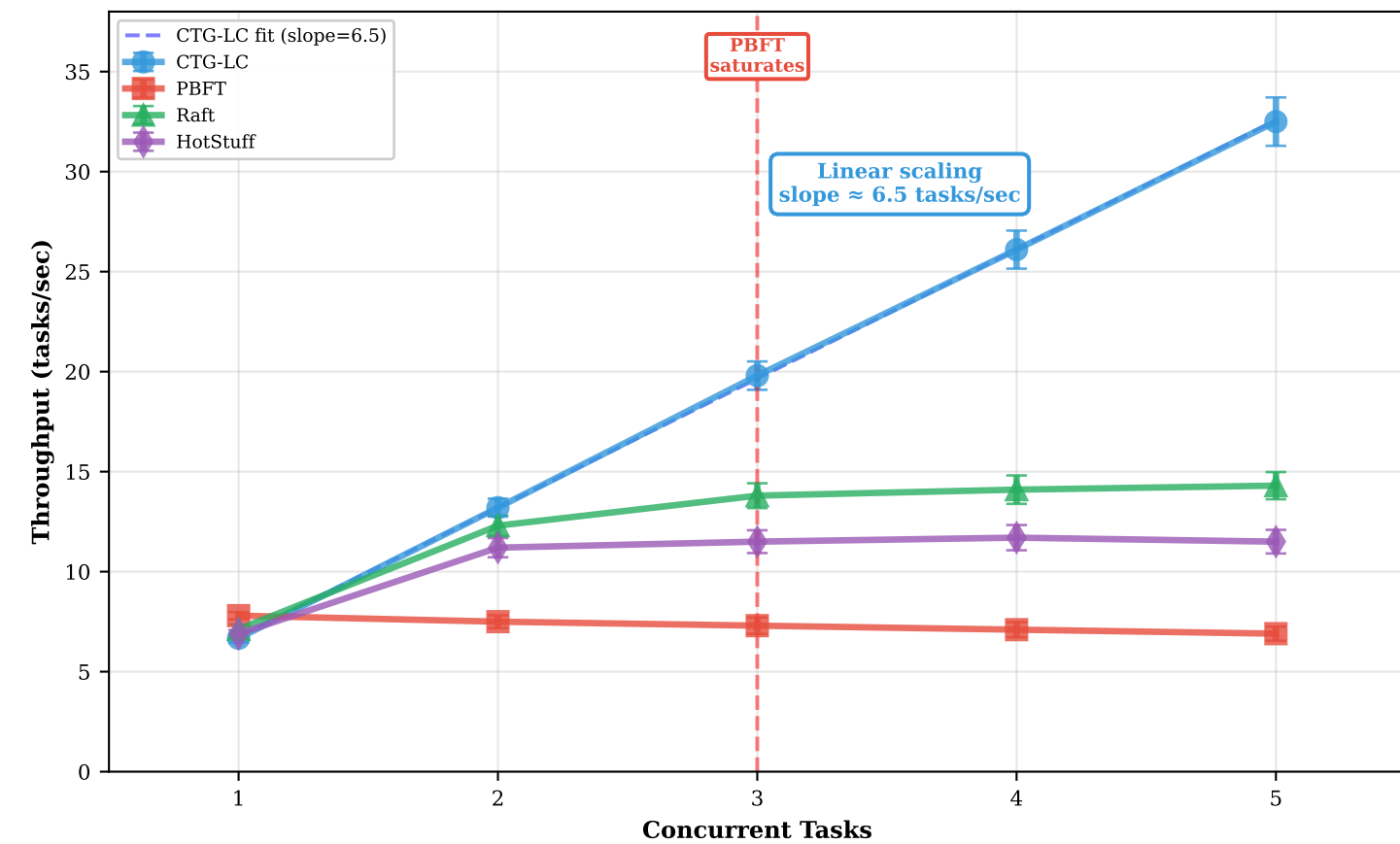
(a) Message Overhead: CTG-LC achieves 98.5% reduction vs. PBFT
(27 vs. 1,770 messages for single task with m=3, k=3, n=30)



(b) Consensus Latency: CTG-LC reduces latency by 60%
(152ms vs 382ms) through spatiotemporal validation (18ms) and localized consensus (134ms)



(c) Throughput: CTG-LC maintains linear scaling (slope ≈ 6.5 tasks/sec) while PBFT saturates at 3 tasks



(d) Network Bandwidth: CTG-LC maintains stable 18% utilization while PBFT exhibits communication storms (>80% peaks)

